

Lecture 2: Review of Probability

Economics 326 — Introduction to Econometrics II

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Randomness

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- **Random experiment:** an experiment the outcome of which cannot be predicted with certainty, even if the experiment is repeated under the same conditions.
- **Event:** a collection of outcomes of a random experiment.
- **Probability:** a function from events to $[0, 1]$ interval.
 - If Ω is a collection of all possible outcomes, $P(\Omega) = 1$.
 - If A is an event, $P(A) \geq 0$.
 - If $A_1, A_2, \dots$ is a sequence of *disjoint* events, $P(A_1 \text{ or } A_2 \text{ or } \dots) = P(A_1) + P(A_2) + \dots$

Random variable

- **Random variable:** a numerical representation of a random experiment.
- **Coin-flipping** example:

Outcome	X	Y	Z
Heads	0	1	-1
Tails	1	0	1

- **Rolling a dice**

Outcome	X	Y
1	1	0
2	2	1
3	3	0
4	4	1
5	5	0
6	6	1

Summation operator

- Let $\{x_i : i = 1, \dots, n\}$ be a sequence of numbers.

$$\sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_n.$$

- For a constant c :

$$\sum_{i=1}^n c = nc.$$

$$\sum_{i=1}^n cx_i = c(x_1 + x_2 + \dots + x_n) = c \sum_{i=1}^n x_i.$$

Summation operator (continued)

- Let $\{y_i : i = 1, \dots, n\}$ be another sequence of numbers, and a, b be two constants:

$$\sum_{i=1}^n (ax_i + by_i) = a \sum_{i=1}^n x_i + b \sum_{i=1}^n y_i.$$

- But:

$$\sum_{i=1}^n x_i y_i \neq \sum_{i=1}^n x_i \sum_{i=1}^n y_i.$$

$$\sum_{i=1}^n \frac{x_i}{y_i} \neq \frac{\sum_{i=1}^n x_i}{\sum_{i=1}^n y_i}.$$

$$\sum_{i=1}^n x_i^2 \neq \left(\sum_{i=1}^n x_i \right)^2.$$

Discrete random variables

We often distinguish between **discrete** and **continuous** random variables.

- A **discrete** random variable takes on only a **finite or countably infinite** number of values.
- The **distribution** of a discrete random variable is a list of all possible values and the probability that each value would occur:

Value	x_1	x_2	$\dots$	x_n
Probability	p_1	p_2	$\dots$	p_n

Here p_i denotes the probability of a random variable X taking on value x_i :

$$p_i = P(X = x_i) \text{ (Probability Mass Function (PMF))}.$$

Each p_i is between 0 and 1, and $\sum_{i=1}^n p_i = 1$.

Example: Bernoulli distribution

- Consider a single trial with two outcomes: “success” (with probability p) or “failure” (with probability $1 - p$).
- Define the random variable:

$$X = \begin{cases} 1 & \text{if success} \\ 0 & \text{if failure} \end{cases}$$

- Then X follows a **Bernoulli distribution**: $X \sim \text{Bernoulli}(p)$.
- PMF:

$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}.$$

Discrete random variables (continued)

- Indicator function:

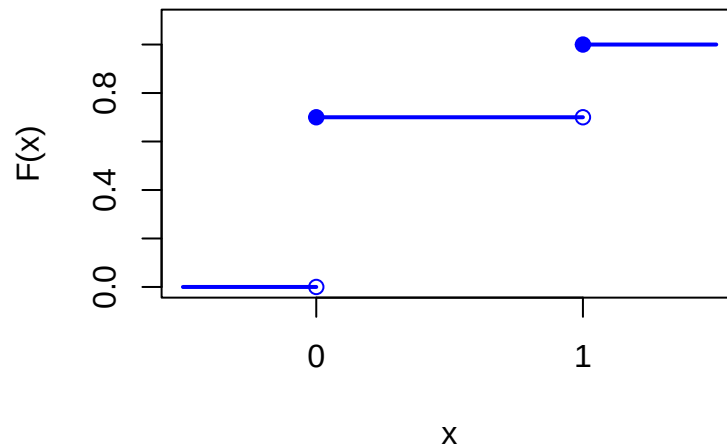
$$\mathbf{1}(x_i \leq x) = \begin{cases} 1 & \text{if } x_i \leq x \\ 0 & \text{if } x_i > x \end{cases}$$

- Cumulative Distribution Function (CDF):**

$$F(x) = P(X \leq x) = \sum_i p_i \mathbf{1}(x_i \leq x).$$

- $F(x)$ is non-decreasing.
- For discrete random variables, the CDF is a step function.

Example: CDF of Bernoulli(0.3)



$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 - p & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

Continuous random variable

- A random variable is continuously distributed if the range of possible values it can take is uncountable infinite (for example, a real line).
- A continuous random variable takes on any real value with **zero** probability.
- For continuous random variables, the CDF is continuous and differentiable.
- The derivative of the CDF is called the **Probability Density Function (PDF)**:

$$f(x) = \frac{dF(x)}{dx} \text{ and } F(x) = \int_{-\infty}^x f(u) du;$$

$$\int_{-\infty}^{\infty} f(x) dx = 1.$$

- Since $F(x)$ is non-decreasing, $f(x) \geq 0$ for all x .

Example: Uniform distribution

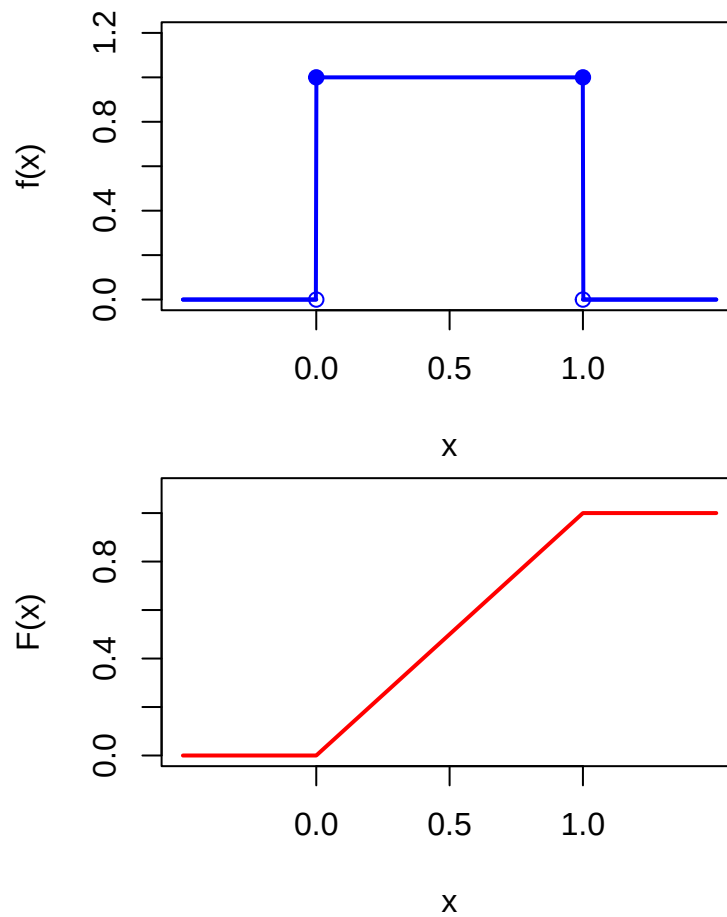
A random variable X follows a **Uniform distribution** on $[0, 1]$, written $X \sim \text{Uniform}(0, 1)$, if it is equally likely to take any value in $[0, 1]$.

PDF:

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{if } x > 1 \end{cases}$$

CDF:

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } 0 \leq x \leq 1 \\ 1 & \text{if } x > 1 \end{cases}$$



Joint distribution (discrete)

- Two random variables X, Y

	y_1	y_2	$\dots$	y_m	Marginal
x_1	p_{11}	p_{12}	$\dots$	p_{1m}	$p_1^X = \sum_{j=1}^m p_{1j}$
x_2	p_{21}	p_{22}	$\dots$	p_{2m}	$p_2^X = \sum_{j=1}^m p_{2j}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_n	p_{n1}	p_{n2}	$\dots$	p_{nm}	$p_n^X = \sum_{j=1}^m p_{nj}$

Joint PMF: $p_{ij} = P(X = x_i, Y = y_j)$.

Marginal PMF: $p_i^X = P(X = x_i) = \sum_{j=1}^m p_{ij}$.

- **Conditional Distribution:** If $P(X = x_1) \neq 0$,

$$p_j^{Y|X=x_1} = P(Y = y_j|X = x_1) = \frac{P(Y = y_j, X = x_1)}{P(X = x_1)} = \frac{p_{1,j}}{p_1^X}$$

Joint distribution (continuous)

- Joint PDF: $f_{X,Y}(x, y)$ and $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$.
- Marginal PDF: $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$.
- Conditional PDF: $f_{Y|X=x}(y|x) = f_{X,Y}(x, y) / f_X(x)$.

Independence

- Two (discrete) random variables are independent if **for all** x, y :

$$P(X = x, Y = y) = P(X = x)P(Y = y).$$

- If independent:

$$P(Y = y|X = x) = \frac{P(X = x, Y = y)}{P(X = x)} = P(Y = y).$$

- Two continuous random variables are independent if **for all** x, y :

$$f_{X,Y}(x, y) = f_X(x)f_Y(y).$$

- If independent, $f_{Y|X}(y|x) = f_Y(y)$ for all x .

Expected value

- Let g be some function:

$$E[g(X)] = \sum_i g(x_i)p_i \text{ (discrete).}$$

$$E[g(X)] = \int g(x)f(x)dx \text{ (continuous).}$$

Expectation is a transformation of a distribution (PMF or PDF) and is a **constant!**

- **Mean** (center of a distribution):

$$E[X] = \sum_i x_i p_i \text{ or } E[X] = \int x f(x) dx.$$

- **Variance** (spread of a distribution): $\text{Var}(X) = E[(X - E[X])^2]$

$$\text{Var}(X) = \sum_i (x_i - E[X])^2 p_i \text{ or } \text{Var}(X) = \int (x - E[X])^2 f(x) dx.$$

- **Standard deviation:** $\sqrt{\text{Var}(X)}$.

Properties

- Linearity:
 - Discrete:

$$\begin{aligned} \mathbb{E}[a + bX] &= \sum_i (a + bx_i)p_i = a \sum_i p_i + b \sum_i x_i p_i \\ &= a + b\mathbb{E}[X]. \end{aligned}$$

- Continuous:

$$\begin{aligned} \mathbb{E}[a + bX] &= \int (a + bx)f(x)dx = a \int f(x)dx + b \int xf(x)dx \\ &= a + b\mathbb{E}[X]. \end{aligned}$$

- Re-centering: a random variable $X - \mathbb{E}[X]$ has mean zero:

$$\mathbb{E}[X - \mathbb{E}[X]] = \mathbb{E}[X] - \mathbb{E}[\mathbb{E}[X]] = \mathbb{E}[X] - \mathbb{E}[X] = 0.$$

Properties (continued)

- Variance formula: $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])] \\ &= \mathbb{E}[(X - \mathbb{E}[X])X - (X - \mathbb{E}[X]) \cdot \mathbb{E}[X]] \\ &= \mathbb{E}[(X - \mathbb{E}[X])X] - \mathbb{E}[(X - \mathbb{E}[X]) \cdot \mathbb{E}[X]] \\ &= \mathbb{E}[X^2 - X \cdot \mathbb{E}[X]] - \mathbb{E}[X] \cdot \mathbb{E}[X - \mathbb{E}[X]] \\ &= \mathbb{E}[X^2] - \mathbb{E}[X] \cdot \mathbb{E}[X] - \mathbb{E}[X] \cdot 0 \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \end{aligned}$$

- If $\mathbb{E}[X] = 0$ then $\text{Var}(X) = \mathbb{E}[X^2]$.

Properties (continued)

- If c is a constant, $\mathbb{E}[c] = c$, and

$$\text{Var}(c) = \mathbb{E}[(c - \mathbb{E}[c])^2] = (c - c)^2 = 0.$$

- $\text{Var}(a + bX) = b^2 \text{Var}(X)$

$$\begin{aligned} \text{Var}(a + bX) &= \mathbb{E}[(a + bX) - \mathbb{E}[a + bX]]^2 \\ &= \mathbb{E}[a + bX - a - b\mathbb{E}[X]]^2 \\ &= \mathbb{E}[bX - b\mathbb{E}[X]]^2 = \mathbb{E}[b^2(X - \mathbb{E}[X])^2] \\ &= b^2 \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= b^2 \text{Var}(X). \end{aligned}$$

- Re-scaling: Let $\text{Var}(X) = \sigma^2$, so the standard deviation is σ :

$$\text{Var}\left(\frac{X}{\sigma}\right) = \frac{1}{\sigma^2} \text{Var}(X) = 1.$$

Example: Bernoulli distribution (continued)

- Recall: $X \sim \text{Bernoulli}(p)$ takes values $\{0, 1\}$ with $P(X = 1) = p$ and $P(X = 0) = 1 - p$.

- Mean:

$$\mathbb{E}[X] = 0 \cdot (1 - p) + 1 \cdot p = p.$$

- Variance:

$$\mathbb{E}[X^2] = 0^2 \cdot (1 - p) + 1^2 \cdot p = p.$$

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1 - p).$$

Example: Uniform distribution (continued)

- Recall: $X \sim \text{Uniform}(0, 1)$ has PDF $f(x) = 1$ for $x \in [0, 1]$.
- Mean:

$$\mathbb{E}[X] = \int_0^1 x \cdot 1 \, dx = \left. \frac{x^2}{2} \right|_0^1 = \frac{1}{2}.$$

- Variance:

$$\mathbb{E}[X^2] = \int_0^1 x^2 \cdot 1 \, dx = \left. \frac{x^3}{3} \right|_0^1 = \frac{1}{3}.$$

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}.$$

Covariance

- **Covariance:** Let X, Y be two random variables.

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

$$\text{Cov}(X, Y) = \sum_i \sum_j (x_i - \mathbb{E}[X])(y_j - \mathbb{E}[Y]) \cdot P(X = x_i, Y = y_j).$$

$$\text{Cov}(X, Y) = \int \int (x - \mathbb{E}[X])(y - \mathbb{E}[Y]) f_{X,Y}(x, y) \, dx \, dy.$$

- $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X] \cdot \mathbb{E}[Y].$$

Properties of covariance

- $\text{Cov}(X, c) = 0.$
- $\text{Cov}(X, X) = \text{Var}(X).$
- $\text{Cov}(X, Y) = \text{Cov}(Y, X).$
- $\text{Cov}(X, Y + Z) = \text{Cov}(X, Y) + \text{Cov}(X, Z).$
- $\text{Cov}(a_1 + b_1X, a_2 + b_2Y) = b_1b_2\text{Cov}(X, Y).$
- If X and Y are independent then $\text{Cov}(X, Y) = 0.$
- $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y) \pm 2\text{Cov}(X, Y).$

Correlation

- Correlation coefficient:

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}.$$

- Cauchy-Schwartz inequality: $|\text{Cov}(X, Y)| \leq \sqrt{\text{Var}(X) \text{Var}(Y)}$ and therefore

$$-1 \leq \text{Corr}(X, Y) \leq 1.$$

- $\text{Corr}(X, Y) = \pm 1 \Leftrightarrow Y = a + bX$.

Conditional expectation

- Suppose you know that $X = x$. You can update your expectation of Y by **conditional expectation**:

$$\mathbb{E}[Y|X = x] = \sum_i y_i P(Y = y_i | X = x) \text{ (discrete)}$$

$$\mathbb{E}[Y|X = x] = \int y f_{Y|X}(y|x) dy \text{ (continuous)}.$$

- $\mathbb{E}[Y|X = x]$ is a constant.
- $\mathbb{E}[Y|X]$ is a function of X and is a **random variable** and a function of X (Uncertainty about X has not been realized yet):

$$\mathbb{E}[Y|X] = \sum_i y_i P(Y = y_i | X) = g(X)$$

$$\mathbb{E}[Y|X] = \int y f_{Y|X}(y|X) dy = g(X),$$

for some function g that depends on PMF (PDF).

Properties of conditional expectation

- Conditional expectations satisfies all properties of unconditional expectation.
- Once you condition on X , you can treat any function of X as a constant:

$$\mathbb{E}[h_1(X) + h_2(X)Y|X] = h_1(X) + h_2(X)\mathbb{E}[Y|X],$$

for any functions h_1 and h_2 .

- Law of Iterated Expectation (LIE):**

$$\mathbb{E}[\mathbb{E}[Y|X]] = \mathbb{E}[Y],$$

$$\mathbb{E}[\mathbb{E}[Y|X, Z] | X] = \mathbb{E}[Y|X].$$

- Conditional variance:

$$\text{Var}(Y|X) = \mathbb{E}[(Y - \mathbb{E}[Y|X])^2 | X].$$

- Mean independence:**

$$\mathbb{E}[Y|X] = \mathbb{E}[Y] = \text{constant}.$$

Example: conditional expectation and LIE

- Joint PMF of $X \in \{0, 1\}$ and $Y \in \{1, 2, 3\}$:

	$Y = 1$	$Y = 2$	$Y = 3$	$P(X = x)$
$X = 0$	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{2}{10}$	$\frac{1}{2}$
$X = 1$	$\frac{2}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{2}$
$P(Y = y)$	$\frac{3}{10}$	$\frac{4}{10}$	$\frac{3}{10}$	1

- Conditional PMFs (divide joint probabilities by marginal of X):

$$P(Y = y|X = 0) : \quad \frac{1/10}{1/2}, \frac{2/10}{1/2}, \frac{2/10}{1/2} = \frac{1}{5}, \frac{2}{5}, \frac{2}{5}$$

$$P(Y = y|X = 1) : \quad \frac{2/10}{1/2}, \frac{2/10}{1/2}, \frac{1/10}{1/2} = \frac{2}{5}, \frac{2}{5}, \frac{1}{5}$$

- X and Y are not independent since the conditional PMFs of Y given $X = x$ are different for $x = 0$ and $x = 1$.
- Conditional expectations: $E[Y | X = x] = \sum_{j=1}^3 y_j P(Y = y_j | X = x)$

$$E[Y|X = 0] = 1 \cdot \frac{1}{5} + 2 \cdot \frac{2}{5} + 3 \cdot \frac{2}{5} = \frac{1+4+6}{5} = \frac{11}{5}$$

$$E[Y|X = 1] = 1 \cdot \frac{2}{5} + 2 \cdot \frac{2}{5} + 3 \cdot \frac{1}{5} = \frac{2+4+3}{5} = \frac{9}{5}$$

- LIE check: $E[E[Y|X]] = \frac{11}{5} \cdot \frac{1}{2} + \frac{9}{5} \cdot \frac{1}{2} = \frac{11}{10} + \frac{9}{10} = 2$
- Direct: $E[Y] = 1 \cdot \frac{3}{10} + 2 \cdot \frac{4}{10} + 3 \cdot \frac{3}{10} = \frac{3+8+9}{10} = 2$

Proof of the Law of Iterated Expectations

- Let X take values $x_1, \dots, x_n$ and Y take values $y_1, \dots, y_m$.

$$\begin{aligned}
 E[E[Y|X]] &= \sum_{i=1}^n E[Y|X = x_i] P(X = x_i) \\
 &= \sum_{i=1}^n \left(\sum_{j=1}^m y_j P(Y = y_j|X = x_i) \right) P(X = x_i) \\
 &= \sum_{i=1}^n \sum_{j=1}^m y_j P(Y = y_j|X = x_i) P(X = x_i) \\
 &= \sum_{j=1}^m y_j \sum_{i=1}^n P(Y = y_j, X = x_i) \\
 &= \sum_{j=1}^m y_j P(Y = y_j) = E[Y]
 \end{aligned}$$

- Key steps:
 - $P(Y = y_j|X = x_i) \cdot P(X = x_i) = P(Y = y_j, X = x_i)$.
 - Changing the order of summation.
 - Summing over i gives marginal $P(Y = y_j)$.

Relationship between different concepts of independence

$$\begin{aligned}
 &X \text{ and } Y \text{ are independent} \\
 &\quad \Downarrow \\
 &E[Y|X] = \text{constant (mean independence)} \\
 &\quad \Downarrow \\
 &\text{Cov}(X, Y) = 0 \text{ (uncorrelatedness)}
 \end{aligned}$$

Normal distribution

- A normal rv is a continuous rv that can take on any value. The PDF of a normal rv X is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \text{ where}$$

$$\mu = E[X] \text{ and } \sigma^2 = \text{Var}(X).$$

We usually write $X \sim N(\mu, \sigma^2)$.

- If $X \sim N(\mu, \sigma^2)$, then $a + bX \sim N(a + b\mu, b^2\sigma^2)$.

Standard Normal distribution

- Standard Normal rv has $\mu = 0$ and $\sigma^2 = 1$. Its PDF is $\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right)$.
- Symmetric around zero (mean): if $Z \sim N(0, 1)$, $P(Z > z) = P(Z < -z)$.
- Thin tails: $P(-1.96 \leq Z \leq 1.96) = 0.95$.
- If $X \sim N(\mu, \sigma^2)$, then $(X - \mu)/\sigma \sim N(0, 1)$.

Bivariate Normal distribution

- X and Y have a bivariate normal distribution if their joint PDF is given by:

$$f(x, y) = \frac{1}{2\pi\sqrt{(1-\rho)^2\sigma_X^2\sigma_Y^2}} \exp\left[-\frac{Q}{2(1-\rho)^2}\right],$$

$$\text{where } Q = \frac{(x-\mu_X)^2}{\sigma_X^2} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} - 2\rho\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y},$$

$$\mu_X = E[X], \mu_Y = E[Y], \sigma_X^2 = \text{Var}(X), \sigma_Y^2 = \text{Var}(Y), \text{ and } \rho = \text{Corr}(X, Y).$$

Properties of Bivariate Normal distribution

If X and Y have a bivariate normal distribution:

- $a + bX + cY \sim N(\mu^*, (\sigma^*)^2)$, where

$$\mu^* = a + b\mu_X + c\mu_Y, \quad (\sigma^*)^2 = b^2\sigma_X^2 + c^2\sigma_Y^2 + 2bc\rho\sigma_X\sigma_Y.$$

- $\text{Cov}(X, Y) = 0 \implies X$ and Y are independent.
- $E[Y|X] = \mu_Y + \frac{\text{Cov}(X, Y)}{\sigma_X^2}(X - \mu_X)$.
- Can be generalized to more than 2 variables (multivariate normal).

Appendix: The Cauchy-Schwartz Inequality

- Claim: $|\text{Cov}(X, Y)| \leq \sqrt{\text{Var}(X)\text{Var}(Y)}$.
- **Proof:** Define

$$U = Y - \beta X,$$

where

$$\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)},$$

- Note that β is a constant!
- Also note the connection to regression and OLS in the definition of β .

- Since variances are always non-negative:

$$\begin{aligned}
0 &\leq \text{Var}(U) \\
&= \text{Var}(Y - \beta X) && \text{(def. of } U\text{)} \\
&= \text{Var}(Y) + \text{Var}(\beta X) - 2\text{Cov}(Y, \beta X) && \text{(prop. of var.)} \\
&= \text{Var}(Y) + \beta^2 \text{Var}(X) - 2\beta \text{Cov}(X, Y) && \text{(prop. of var., cov.)} \\
&= \text{Var}(Y) + \underbrace{\left(\frac{\text{Cov}(X, Y)}{\text{Var}(X)} \right)^2}_{=\beta^2} \text{Var}(X) \\
&\quad - 2 \underbrace{\left(\frac{\text{Cov}(X, Y)}{\text{Var}(X)} \right)}_{=\beta} \text{Cov}(X, Y) && \text{(def. of } \beta\text{)} \\
&= \text{Var}(Y) + \frac{\text{Cov}(X, Y)^2}{\text{Var}(X)} - 2 \frac{\text{Cov}(X, Y)^2}{\text{Var}(X)} \\
&= \text{Var}(Y) - \frac{\text{Cov}(X, Y)^2}{\text{Var}(X)}.
\end{aligned}$$

- Rearranging:

$$\text{Cov}(X, Y)^2 \leq \text{Var}(X) \text{Var}(Y)$$

- or

$$|\text{Cov}(X, Y)| \leq \sqrt{\text{Var}(X) \text{Var}(Y)}.$$